

● PROJECT OVERVIEW & THESIS

AgroPulse: *AI-Based Precision Fertilizer Recommendation*

A smart farming system that combines a soil moisture sensor with Google Gemini Vision AI to tell farmers exactly how much water and fertilizer to use.

⚡ WHAT THE PROJECT DOES

AgroPulse is an AI-powered precision farming platform that combines real-time soil data from IoT sensors with advanced AI from **Google Gemini** and **Grok Vision** trained on massive multimodal datasets. It provides ICAR-calibrated fertilizer recommendations in safe, practical doses that farmers can use directly in the field.

PART 1 • THE SENSOR

Checking Soil Moisture

A hardware probe placed in the soil measures water percentage and air temperature in real time, so we know if the ground is dry, healthy, or waterlogged.

PART 2 • THE AI VISION

Scanning the Leaf

A phone camera photo is analyzed by Google Gemini AI to identify the plant type and spot visible problems like yellow leaf edges, brown dead spots, or fungal mold.

PART 3 • THE RECIPE

The Exact Fertilizer Recipe

Translates agricultural formulas into safe, everyday measurements—how many buckets of water and how many spoonfuls of fertilizer—so farmers never burn their crops.

● THE CORE PROBLEM

The Deadly Guesswork: *Why Yellow Leaves Trick Farmers*

Looking at leaf color alone is dangerous. Two plants can look identical on top, but have completely opposite problems in the soil.

THE DEADLY PARADOX

Two Leaves Look Identical. One Needs Water, The Other Will Drown.

CASE 1 • THIRSTY PLANT (DRY SOIL: 18%)

Yellow Margins + Dry Ground

The roots lack water, making leaves turn yellow. The plant needs **water first** so fertilizer can be safely dissolved without burning the roots.

CASE 2 • DROWNING PLANT (SOAKED SOIL: 82%)

Yellow Margins + Waterlogged Ground

Roots are suffocating in mud, so they cannot absorb nutrients. Adding chemical fertilizer will **burn roots and kill the plant within 48 hours**.



Camera-only phone apps only look at the yellow leaf and tell farmers to add chemicals—accidentally killing drowning plants.

68%

Chemical Over-Dosing Rate

Farmers over-apply synthetic Urea due to panic and guesswork, wasting money on unneeded chemicals.

18 Days

Soil Testing Lab Delay

Government soil testing laboratories take weeks to return results—far too slow for sick plants.

4.4 pH

Soil Acidification Danger

Dumping excess chemical fertilizer destroys good soil microbes and damages long-term crop yield.

• OUR SOLUTION

The Solution: *Checking the Leaf & the Soil Together*

Photo-only apps fail. AgroPulse cross-checks what the camera sees on the leaf with what the sensor measures in the soil.

ABOVE GROUND • THE CAMERA'S JOB

What Google Gemini AI Sees

- ✓ **Identifies the Plant**
Recognizes Tomato, Potato, Maize, Chilli, Mango, and other regional crops automatically.
- ✓ **Spots Leaf Problems**
Detects yellowing edges, brown dead spots, powdery white mold, and leaf rot.
- ✓ **Rejects Invalid Photos**
Automatically blocks photos of faces, shoes, pets, or indoor objects to keep advice accurate.

BELOW GROUND • THE SENSOR'S JOB

What the Soil Sensor Measures

- ✓ **Soil Moisture Percentage (%)**
Directly reads how wet or dry the soil is around the root zone.
- ✓ **Air Temperature & Humidity**
Checks temperature (°C) and air moisture (% RH) with an integrated DHT11 sensor.
- ✓ **Fungus Risk Warning**
Warns if muggy, humid weather will cause mold to grow after watering.

THE GOLDEN RULE

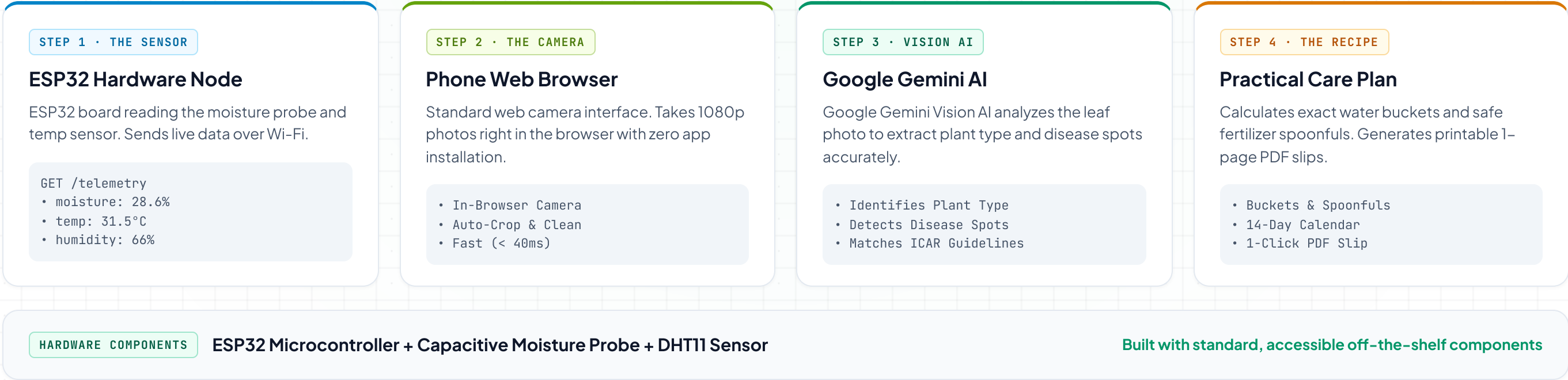
How the System Decides:

If soil is dry (< 25%) → **Water First, Then Feed** | If soil is soaked (> 70%) → **Stop Watering, Don't Add Fertilizer**

● SYSTEM PIPELINE

How We Built It: *From the Sensor to the Screen*

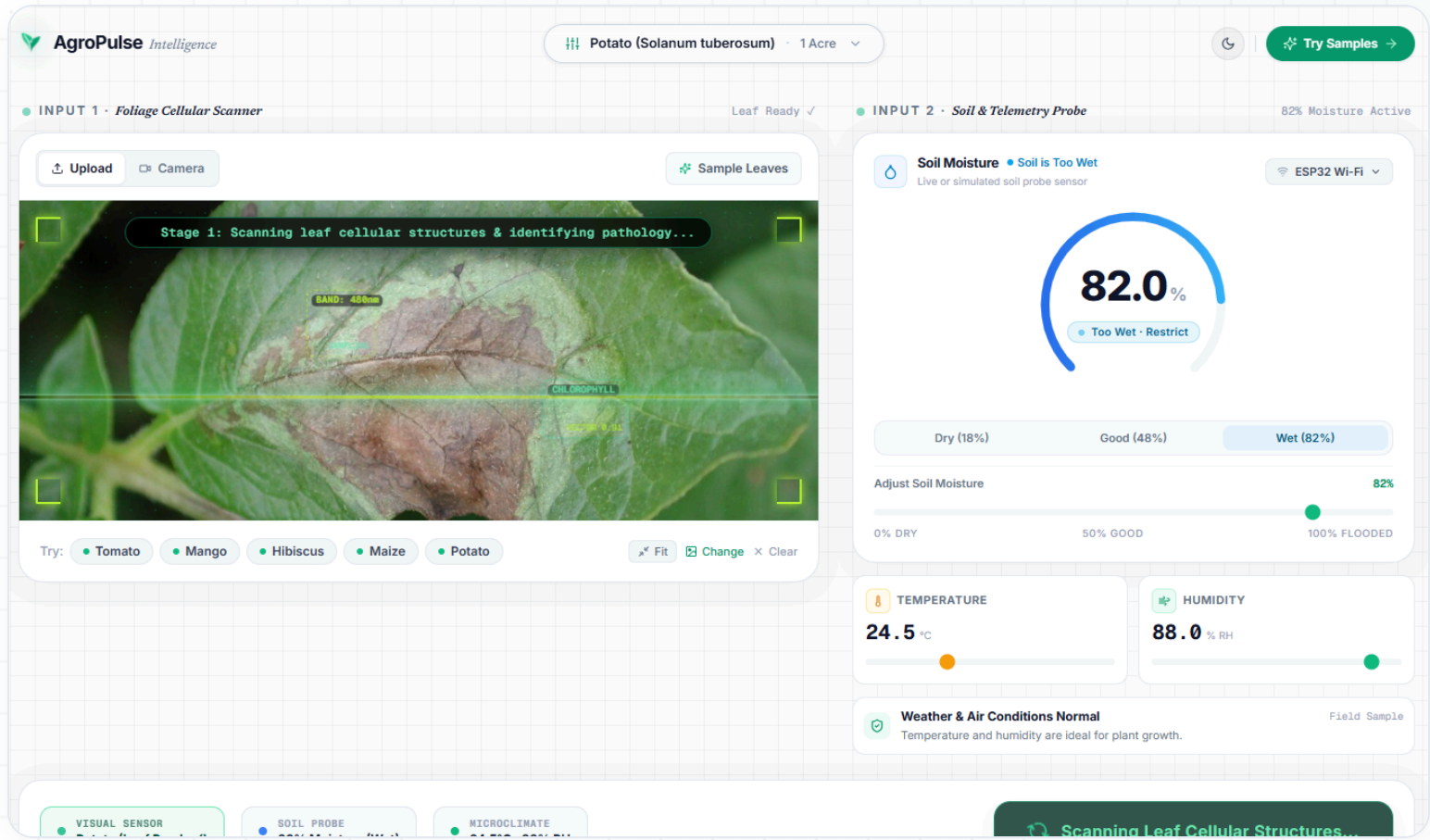
How the hardware probe, phone camera, cloud AI, and recipe calculator work together step-by-step.



● LIVE APP WALKTHROUGH

Step 1 in the App: *Framing & Scanning the Leaf*

A clean, simple camera screen designed for farmers to use outdoors in bright sunlight.



FEATURE 1 · EASY FOCUS GUIDE

Target Box for Easy Framing

A clear green guide box helps the farmer center the sick leaf and hold steady for a sharp picture.

FEATURE 2 · SIMPLE ENGLISH

No Confusing Botanical Terms

Instead of "Chlorosis" → "Yellow leaf edges"

Instead of "Necrotic lesions" → "Dead brown spots"

Instead of "Rot lesions" → "Dark wet rotting patches"

FEATURE 3 · INSTANT FEEDBACK

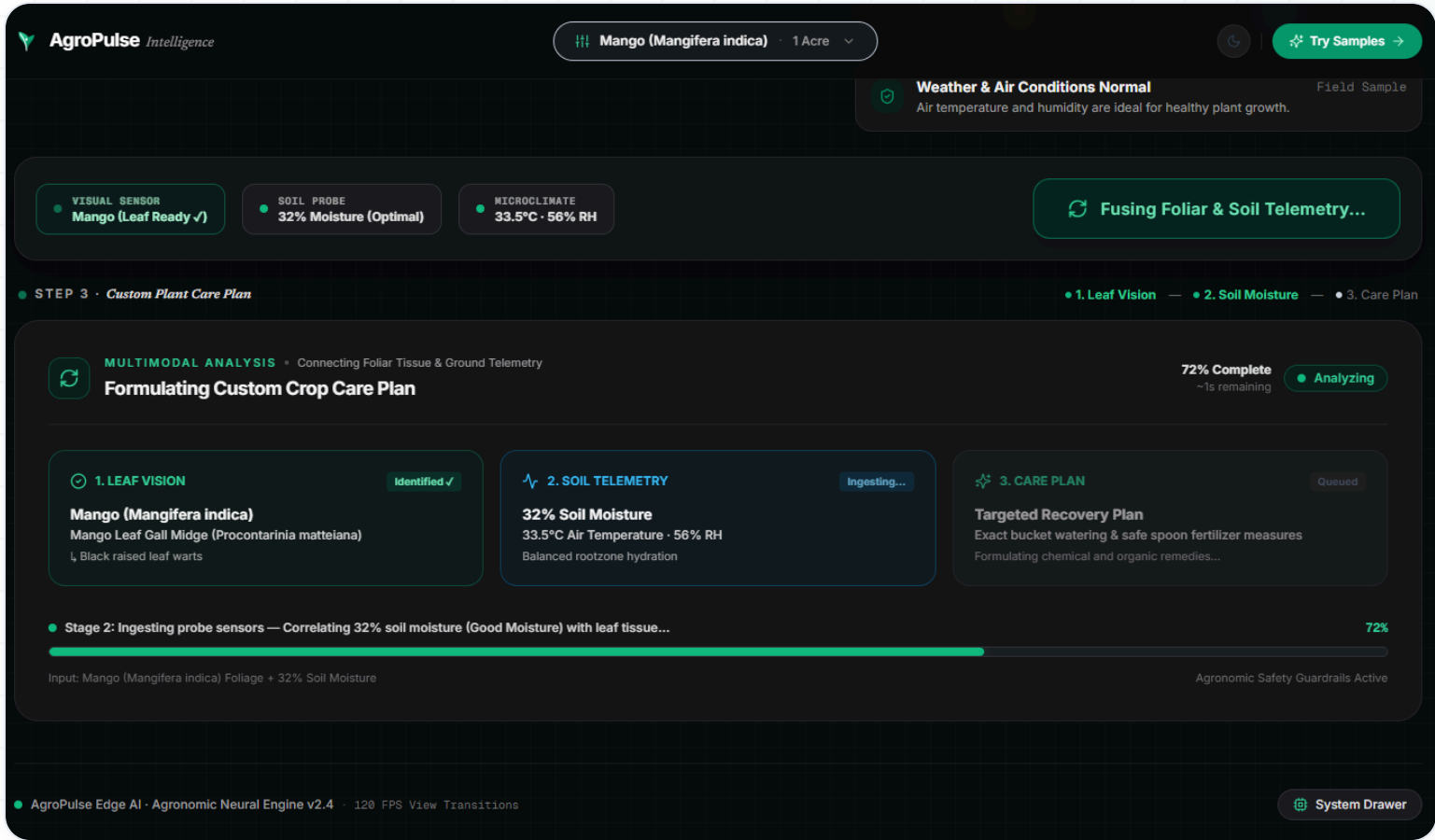
Visual Scan Confirmation

A smooth scan line shows the farmer that the picture has been captured and is being analyzed by Gemini AI.

● LIVE APP WALKTHROUGH

Step 2 in the App: *Combining Sensor Numbers with the Photo*

A clear 3–step progress bar showing the farmer exactly how their soil readings adjust the advice.



STEP 1

COMPLETED ✓

Identifying the Plant & Disease

Gemini AI identifies the crop (Potato) and detects early fungal blight spots on the leaf.

STEP 2

ACTIVE CHECK

Reading Soil Moisture

Checks live sensor numbers (28.6% moisture – dry soil, 31.5°C temp) to see how thirsty the roots are.

STEP 3

FINALIZING

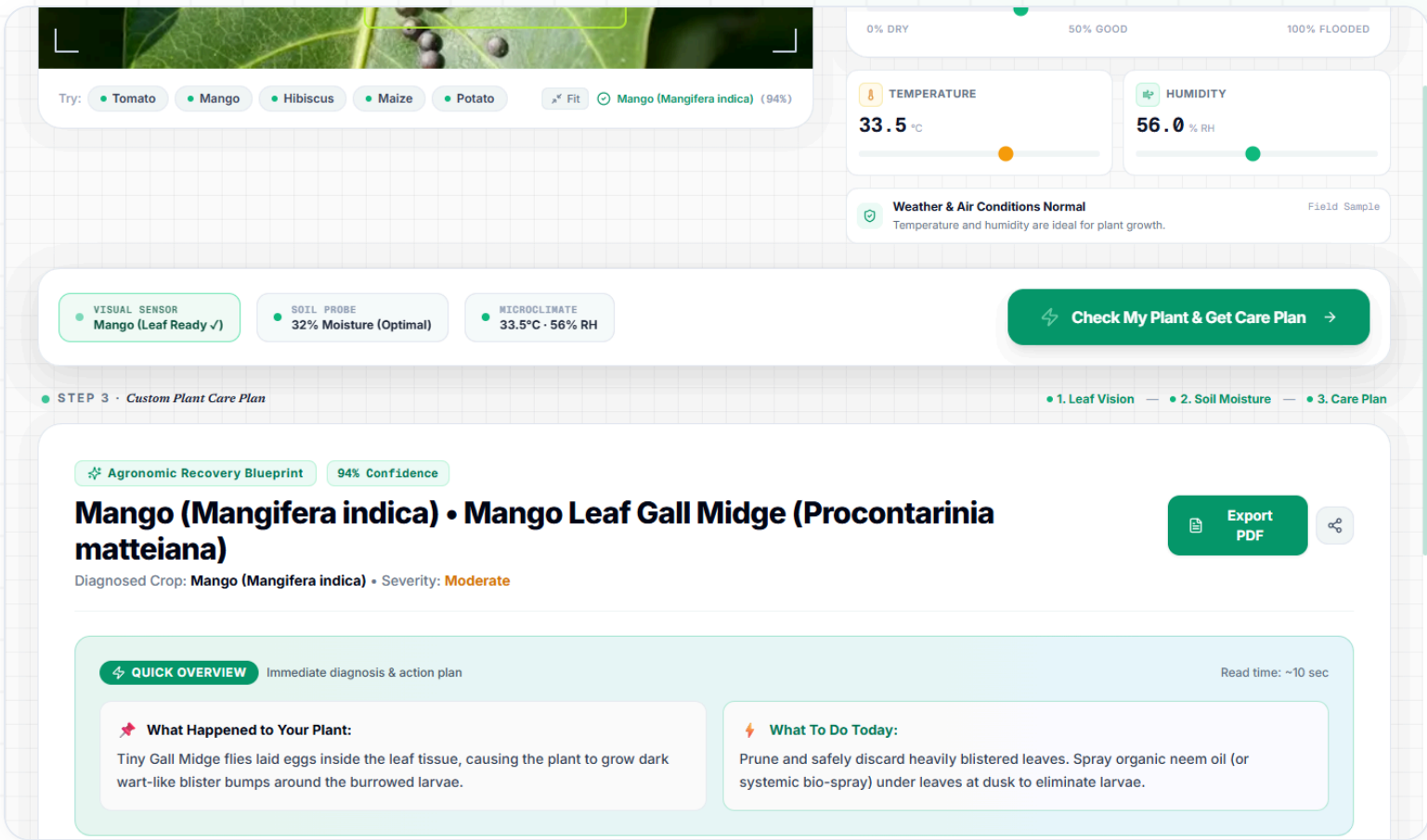
Calculating the Safe Dose

Because the soil is dry, the system automatically adjusts the water amount so fertilizer won't burn the roots.

● LIVE APP WALKTHROUGH

Step 3 in the App: *The Practical Fertilizer Recipe*

No confusing farm formulas like kilograms-per-hectare. Everything is in simple buckets and spoonfuls.



⚡ 30-SECOND SUMMARY

What Happened & What To Do Today

Gives the farmer a quick 30-second summary: why soil conditions caused the issue and the single most important action for this evening.

💧 WATER FIRST

How Much Water to Give

Specifies exact bucket count (e.g. "Apply 1 standard 15L bucket around rootzone") to prevent fertilizer burning dry roots.

🥕 TWO RECIPE CHOICES

Chemical & Organic Options

Toggle between chemical NPK (1 level tablespoon per 10L bucket) and organic compost tea (2 cups vermicompost). Zero guesswork.

• EVALUATION & RESULTS

Testing Results & *How It Works Offline*

Sub-3.2 second speed, affordable hardware, works even without internet, and future plans for larger fields.

< 3.2s

Fast Response Time

From taking the photo to getting the recipe on screen.

3-Part

Hardware Sensor Stack

ESP32 board + soil moisture probe + DHT11 sensor.

2-Tier

Works Without Internet

Cloud AI online → Local sensor triage offline.

-42%

Fertilizer Waste Saved

Stops chemical over-dosing and protects soil health.

OFFLINE SAFETY • 2 WAYS IT RUNS

How AgroPulse Works in Remote Fields (No Internet)

Mode 1 • Connected Mode (Cloud AI): When internet is available, leaf photos are analyzed by Google Gemini AI for complete disease identification.

Mode 2 • Remote Field Mode (Offline Sensor): When there is no internet, the phone connects directly to the ESP32 sensor. A built-in offline rules engine reads soil moisture and gives instant emergency watering advice with zero cloud connection.

FUTURE PLANS • 2026 - 2027

PHASE 1 • CURRENT PROTOTYPE

Working Web Platform

ESP32 soil telemetry, camera interface, Google Gemini AI, dual recipes, and printable PDF care slip.

PHASE 2 • LATE 2026

Long-Range LoRa Radio

Connect multiple sensors across a 10-acre farm up to 5km away without needing cellular data.

PHASE 3 • 2027

Automatic Drip Dispensing

Solar-powered water valves that release the exact fertilizer mix directly through drip irrigation tubes.